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Where Minds Pollinate

Computer Science



Class 9

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Chapter 5

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CHAPTER#

05

"APPLICATION OF COMPUTER"

Topic: 5.1 (A.I & M.L)

Q- Differentiate between AI & M.L?

ARTIFICIAL INTELLIGENCE

MACHINE LEARNING

- | | |
|--|--|
| <ul style="list-style-type: none">• It is branch of computer science.• It work like a human brain.• It uses different machines models in robots or AI machine.• e.g Robots working as waiter, room cleaner. | <ul style="list-style-type: none">• It is subset of Artificial Intelligence.• It trains machines from past data inputs.• M.L models can perform work like image & voice recognition, identifying spam messages.• e.g Self-driving cars, google assistant etc. |
|--|--|

Q2- Define Types of ML?

- Answer (i) Supervised Learning.
(ii) Un-supervised Learning.

SUPERVISED LEARNING

Definition:

It involves training a model on labeled data, where correct output is already known.

Key Characteristics:

- Label data.
- Correct output is known.
- Model learn from example.

Example:

- image classification.
- prediction of car price.

UN-SUPERVISED LEARNING

Definition:

It involves training a model on unlabeled data, where correct output is known.

Key Characteristics:

- Unlabel data.
- No correct output is known.
- Model discover hidden pattern.

Examples:

- Bank need to check an applicant to whether give loan or not.

SLO'S

Q- Write / Explain the process of machine Language for training a model?

- Answer (i) Collection of training data.
- (ii) Creating algorithms.
- (iii) Learning process.
- (iv) Creating training model.
- (v) Predicting results.

Exercise Questions:-

Q- Briefly describe the positive impact, in our daily life?

- Answer (i) AI solve complex problem.
(ii) AI helps in designing & production of high quantity products.
(iii) AI system used in hospital to provide better diagnosis health diseases.
(iv) Help in travelling by providing shortest routes.
(v) Help in improving market sales.

SLO'S

Q- What are negative impact of AI system?

- Answer (i) Drivers less self-driving cars can have an accident.
(ii) Error is AI health system can damage human life.
(iii) AI based automation can lead to job losses.
(iv) Human loss creativity & critical thinking because of AI tools.
(v) AI system can be biased.

09-10-2025

Classwork

Thursday

Topic # 5.1 (AREAS OF A.I)

Q-. Differentiate between Areas of A.I?

Answer (1)

SPEECH RECOGNITION:

- Convert voices into text forms.
- Speech Recognitions used in virtual assistants like Siri, Alexa, and Cortana.

LIMITATIONS:

- It cannot recognise different pronunciation.
- Due to lack of accuracy in speech cause results.

(2)

COMPUTER VISION:

- Computer vision is used in traffic controls.
- For home security.
- Provide live update by images & videos.
- e.g: Self-driving car, facial recognition etc.

(3)

NLP (Natural Language Processing):

- It is branch of AI.
- It is used in software when we need to recognize natural language in the spoken or written.
- Used in email filtering language translation.

(4) EXPERT SYSTEM:

- Those systems which can perform task without any help of humans.
- e.g: Diagnoses diseases in the absence of doctor.
- Expert system can help doctor in surgeries.
- Fraud Detection.
- Planning & scheduling projects.

SLO's

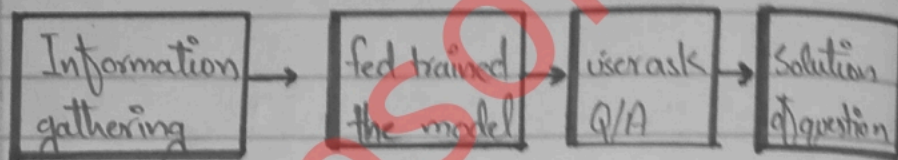
Q - Write down the steps of development of Expert system?

Answer (1) Gather knowledge about a specific area is obtained from experts & fed in the expert system. A knowledge databar is created.

(2) When end user makes a query, the expert system asks the user to enter required information about it.

(3) The expert system analyze the query. & applies rules & reasoning to extract knowledge from its databar.

(4) Finally, the expert system will provide the solution to end user.



Q - What is Cloud computing?

Answer Cloud computing refers to provide computing resources via internet like server, storage, databox etc.

SERVER PROVIDED BY CLOUD COMPUTING:

- 1) Servers
- 2) Databox
- 3) Storage
- 4) Networking
- 5) Software

Q- What are the benefits of cloud computing?

Answer(1) Scalability:

Scale up or down as needed, without huge upfront costs.

(2) Cost-Efficiency:

Pay only for what you use.

(3) Accessibility:

Access data & apps from any where on any device, with internet.

(4) Reliability:

Cloud provides ensures high uptime & minimize down time risks.

(5) Automatic Updates:

It provides handle updates & maintenance.

Q- Differentiate between public & private cloud?

Public Cloud	Private Cloud
- Resources owned & operated by a cloud service provider. - Delivered over the internet.	- Resources owned by & used by a single organization. - It is not used by public.

Managed & maintained by
~~maintain~~ service provider.

Amazon web services,
microsoft azure.

- Managed & maintained
by the organization.

- Using by organizations
needed security & control
like forces, banks etc.